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Improving IoT security through an explainable hybrid CNN-transformer model and federated learning

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The rapid proliferation of Internet of Things (IoT) devices has intensified cybersecurity threats, exposing critical infrastructure to sophisticated intrusion attacks. Existing intrusion detection systems (IDS) typically rely on centralized architectures that compromise data privacy, or employ single-architecture models that fail to capture both local spatial patterns and long-range temporal dependencies in network traffic simultaneously. To address these limitations, this paper proposes a novel explainable hybrid CNN-Transformer model integrated with federated learning (FL) for privacy-preserving intrusion detection in IoT environments. The proposed framework uniquely combines four key components not previously integrated in this context: a dual-block CNN-Transformer architecture, federated learning with FedAvg aggregation, multi-class attack classification, and Local Interpretable Model-Agnostic Explanations (LIME) for decision transparency. Evaluated on the IoT-23 dataset across both federated and non-federated scenarios, the proposed model achieves 94.89% accuracy in federated binary classification and 92.17% in federated multi-class classification, outperforming standalone CNN and ensemble baselines by significant margins. Generalizability is further validated through an ablation study on the CIC IoT-DIAD 2024 dataset. The integration of LIME provides actionable feature-level explanations that support real-time decision-making for network security analysts, advancing both the interpretability and trustworthiness of automated IoT intrusion detection systems.

Keywords IoT security, Intrusion detection, CNN-transformer, Federated learning, Centralized learning, Explainable AI (XAI)

The Internet of Things (IoT) zone has seen great development characterized by a large number of packages, thereby producing an increasing array of enterprise potentialities in recent years. As an essential element of the approaching era of the net, IoT is widely applied across numerous contexts. However, it is no longer limited to smart cities, clever homes, and industrial IoT to facilitate statistics-driven decision-making and sensible automation through profound engagement with the physical environment[1,2]. Despite its super achievements, the total capability of the IoT has yet to be found out, especially because of its insecurities. The demanding security situations of IoT are derived from numerous elements, including the growing amount of data, the growing range of devices, and the embedded nature of IoT devices[3,4]. These elements heighten the heterogeneity, the limited computation capability, and the confined energy, storage, and bandwidth of many IoT devices[5]. Threats like Distributed Denial of Service (DDoS) assaults, botnets, and data breaches are increasingly targeting IoT networks, underscoring the need for a strong IDS able to correctly identify and mitigate these threats[6,7]. These demanding situations are leading to further safety dangers and capability vulnerabilities and are affecting many

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fields, which include privacy troubles, infrastructure protection, and the security of diverse embedded software and system offerings.

Over the past decade, numerous contributions have been made regarding the protection against security issues and the efficient detection of malicious attacks. To improve the performance of the attack detection process, the most traditional centralized mechanism is to gather the data from an IoT network environment and then transmit it to an external server for the detection of malicious activities by AI-powered models locally[8].

The AI-powered models have exhibited strong capabilities in detecting malicious activities, and their training processes have a strong demand for large-scale and diverse datasets[9]. However, IoT devices are often equipped with low-capacity hardware, computing, storage, and power resources, which may limit their contribution to AI model training[10]. Moreover, considering the problem of privacy preservation, the communication cost needed to transmit large amounts of data may be intolerable, especially when the data are sensitive and require effective protection. Advances in machine learning and deep learning have the potential to improve security for IoT[11]. However, traditional methods that rely on centralized data storage and processing raise privacy concerns and are not easily scalable across distributed IoT networks. Federated learning is a method of machine learning in which a model is trained using separate local models from multiple decentralized clients. Only model updates from the clients are exchanged between clients and aggregations to the global server, and no raw data is exchanged across the systems[12]. FL is particularly useful in privacy-sensitive device data settings, where user data is not allowed to leave the user's device[13]. Despite these benefits, challenges with accuracy and understanding complex IoT datasets still need to be addressed. In parallel, hybrid models, specifically those combining CNN and transformer architectures, have shown potential in addressing the unique data characteristics of IoT environments. CNNs excel in capturing spatial patterns within network data, while transformers effectively model temporal dependencies and long-range interactions[14]. By merging these strengths, a transformer-CNN hybrid model can significantly improve the accuracy of intrusion detection systems, particularly in multi-class classification tasks involving various types of IoT-based attacks.

Despite these advancements, existing approaches often focus solely on centralised learning or lack effective hybrid models for capturing complex patterns in IoT data. Moreover, many models fail to incorporate explainability, which is crucial for gaining user trust and ensuring transparency in cybersecurity decisions.

To address the gaps in detection accuracy and transparency, this paper proposes an explainable hybrid CNN transformer-based model within a non-federated and federated learning framework scenario for IoT attack detection using the IoT-23 dataset. The model is designed to improve attack classification accuracy while incorporating explainability through LIME, allowing users to understand model predictions and build trust in the system's decision-making process. This study evaluates the model's performance in both non-federated and federated learning settings, demonstrating its effectiveness in accurately detecting diverse IoT attacks while preserving data privacy. The contributions of this paper are as follows:

- Propose a dual-block CNN-Transformer architecture that simultaneously captures local spatial features via convolutional layers and long-range temporal dependencies via multi-head self-attention for both binary and multi-class classification.
- Integrate the proposed hybrid model within a four-client federated learning framework using FedAvg aggregation, demonstrating that privacy-preserving decentralised training achieves 94.89% binary classification accuracy.
- Incorporate LIME-based explainability (XAI-FL) to generate per-prediction feature attributions, enabling network security analysts to identify which traffic features drive attack classification decisions.
- Validate model generalizability through a comprehensive ablation study on the CIC IoT-DIAD 2024 dataset, confirming that the proposed framework maintains robust performance across diverse IoT attack.

The rest of this paper is structured as follows: section "Literature review" covers the literature review, which examines the attacks of IoT in non-federated and federated scenarios, as well as the outcomes of AI models. Section "Methodology" introduces the proposed hybrid explainable CNN transformer-base model and research methods. Section "Results" displays the discussed experimental findings. Lastly, section "Discussion" introduces the conclusion and future work of the paper.

Literature review

In this section, we will be conducting a comprehensive and extensive comparison between the previously conducted studies, diligently examining and thoroughly exploring the various facets of how machine learning models are efficiently deployed in IoT attack detection and put to use in both non-federated and federated learning scenarios.

IoT security and attack detection in a non-federated learning approach

The security of Internet of Things (IoT) systems has been a topic of extensive research, as there is a growing worry about the risks and dangers that interconnected devices face. The IoT and ML have seen significant growth in sectors such as healthcare, manufacturing, agriculture, and transportation, with applications ranging from sales prediction to device monitoring and security improvement[15]. In this section, we examine the current body of literature that focuses on IoT attack datasets in scenarios involving both non-federated learning with central data training and federated learning. For example, the studies[16,17] have put forward a deep learning methodology for identifying irregularities in IoT network traffic, which has demonstrated remarkable precision in detecting malicious behaviors. In the same way[18], used machine learning methods to forecast and stop vulnerabilities in IoT devices by examining how they operate and recognizing possible patterns for attacks. Umair et al.[19] explored IoT security using the N-BaIoT dataset, revealing the potential of spiking neural networks for malware

classification on IoT devices. Similarly, Okur et al.[20] highlighted the critical role of robust intrusion detection systems in defending against DDoS attacks, underscoring the urgent need for effective detection techniques to counter these significant threats.

The IoT-23 dataset was employed by the authors to explore the diverse range of threats present in IoT networks. For example, the authors in this study emphasize the importance of uniform features, providing a fundamental basis for future studies on detecting intrusions[21]. Using the open-source Catboost for IoT security, the authors of[22] presented an improved intrusion detection system that was based on gradient boosting (GB) and decision trees (DT). NSL-KDD, IoT-23, BoT-IoT. The edge-IoT is one of the improved datasets that have been utilized to improve the architecture of the experiment. In addition, B. Roy et al.[23] introduced a model for detecting IoT attacks using a bidirectional long short-term memory, which is a type of recurrent neural network (RNN). The model has shown an accuracy of 95%, but it was trained with a small sample of only 5451 test samples from one dataset. Also, when applied to a multi-class in IoT-23 datasets, only 72% accuracy was reached. Additionally, the authors did not compare their model with other current models.

In the study[24] which had tested four different algorithms on the IoT-23 dataset: CNN, Decision Trees, Naive Bayes, and SVM. In this study, all labels in the IoT-23 dataset were examined, and the highest accuracy of 73% was achieved by the decision tree model. Additionally, CNN achieved 69.35% accuracy, and SVM achieved 69% accuracy. Alhamad, R. N. & Alserhani, F. M also analyzed all labels in the same dataset using various models such as Decision Trees, RF, Naive Bayes, SVM, AdaBoost, XGBoost, CNN, and Multi-Layer Perceptron. The highest accuracy achieved in this analysis was 74% using decision trees, RF, and MLP models. Complementing this, try to enhance the prediction of all types of IoT-23 attacks, as in the experiment in the study[25] aimed to identify all labels, excluding the Okiru malware. The models performed with an accuracy rate of 89%. The performance of boosting strategies was compared with the performance of ensemble-based techniques that did not use boosting, as described by Garg et al.[26]. These two forms of attacks were identified as binary class and multiclass output, respectively: IoT attacks and DDoS attacks. BoT–IoT, IoT-23, and CIC–DDoS-2019 were the three data sets that were utilized for the review process. Both XGB and LGB, which are both techniques of boosting, were utilized. The LGB demonstrated the highest level of performance, achieving an accuracy of 94.79%.

Furthermore, A. Nazir et al.[27] proposed a combination CNN-LSTM model for detecting threats in IoT. This model uses data from different datasets, such as IoT-23, N-BaIoT, and CICIDS2017, and can classify 14 types of attacks. They achieved an impressive accuracy of 95% on the IoT-23 dataset. In recent years, there have been significant advancements in the security of IoT networks. More researchers are now focusing on ensemble learning as a highly effective method to enhance the performance of intrusion detection systems[28–30]. In the initial research conducted on the IoT-23 dataset, the main emphasis was on integrating conventional machine-learning techniques to enhance the detection of cyberattacks.

IoT security and attack detection with hybrid, ensemble, and transformer models in the federated learning approach

The studies mentioned in the previous section have consistently applied threat testing to machine learning for its effectiveness in identifying and mitigating potential attacks. However, one common challenge in these studies is their application to centralized structures, which often require the collaborative training necessary to gather comprehensive threat intelligence[27]. As a result, there has been growing interest in leveraging blockchain technology as an emerging solution for federated management of threat intelligence statistics[31]. The decentralized and immutable nature of blockchain technology allows for overcoming the limitations of traditional methods, providing a platform for improved collaboration, data sharing, and consensus in threat intelligence efforts[32].

An alternative approach to improving detection accuracy was used: hybrid models that combine multiple recognition techniques, such as CNN-RNN, to improve test accuracy[33]. This approach leveraged spatial and temporal statistical patterns in attacks on the IoT dataset 23. This hybrid model demonstrated promising results, achieving an accuracy of approximately 93% for binary classes and 90% for multi-class classes. However, due to their complexity and resource requirements, these hybrid models may be less suitable for deployment on IoT devices with limited assistance.

In recent years, federated learning has emerged as a method for protecting privacy in IoT security. The federated learning model achieved strong performance across all phases of evaluation. Confusion matrices illustrated high accuracy with minimal misclassifications for various intrusion types, such as DDoS, backdoor, injection, and reconnaissance[34]. This highlights the model's effectiveness in distinguishing between different types of attacks. Study[35], utilized federated learning with a mix of CNN and LSTM models on the IoT-23 dataset, producing strong results in a decentralized learning scenario. Their method upheld privacy by training the models on edge devices without centralizing data, which is important for IoT setups. However, the accuracy was a little lower, approximately 88–90%, compared to centralized models due to the challenges of distributed learning. Sorour et al. (2025) proposed an LSTM-JSO framework for privacy-preserving, adaptive intrusion detection in federated IoT networks[36]. They introduced a Joint Strategy Optimization (JSO) algorithm to optimize hyperparameters dynamically during training, significantly improving model adaptability and performance. Experimental results on multiple real-world datasets (including TON_IoT) demonstrated that the proposed LSTM-JSO model achieved superior accuracy, precision, recall, and F1-score compared to existing methods. Furthermore, due to increasing privacy issues, several research studies have explored the use of federated learning along with ensemble models for enhancing security in IoT attack detection[37]. A hybrid ensemble model that combines CNN, LSTM, and XGBoost was introduced in a federated learning framework[38]. The model was trained on IoT devices locally and combined globally, ensuring privacy while delivering high performance. This method demonstrates potential, especially in situations where centralizing data is not feasible.

Transformer models, initially created for processing natural language, have now been utilized in the field of network security because of their capacity to comprehend intricate connections[39]. By incorporating self-

attention mechanisms[40], it can effectively capture the complex interdependencies of sequential network data features, resulting in the successful identification of IoT attacks. Despite this achievement, it was discovered that using transformers alone was demanding on resources and could lead to overfitting when working with smaller datasets. As a solution, a hybrid approach was developed that combines CNNs for the initial extraction of features with transformers for sequence modeling. In addition, the integration of CNNs and transformers has proven to be an effective method for improving the detection of attacks in IoT systems. One such example is a hybrid model proposed by researchers in[41], which combines CNN layers to extract spatial features from network data with transformer layers to capture temporal dependencies. This model demonstrated enhanced accuracy in both binary and multi-class classification when compared to using CNN or Transformer models alone. The hybrid architecture effectively balances the efficiency of CNNs with the contextual understanding of transformers, making it particularly suitable for complex multi-class classification in IoT-23 scenarios. Recent efforts have aimed to make explainable AI models more interpretable due to the perception of these models as black boxes[42]. However, there is a lack of research exploring explainable AI techniques on the IoT-23 dataset. For instance, the integration of explainable AI methods such as SHAP or LIME with AI models has been used to elucidate feature contributions in identifying certain attacks[43]. This is essential for allowing cybersecurity experts to comprehend model decisions and have confidence in their results, but it still requires additional research and improvement. Table 1a presents an overall comparison of all previous studies above that have related to IoT intrusion attacks.

Table 2b presents a structured comparison of the proposed framework against six closely related studies drawn from the surveyed literature. Ullah et al.[10] and Yao et al.[41] both employ CNN-Transformer hybrid architectures but without federated learning or explainability components. Roy[18] and Alhamad & Alserhani[20] evaluate on the IoT-23 dataset but are restricted to single architectures without FL or XAI, with Roy achieving only 72% multi-class accuracy. Torre et al.[34] and Sorour et al.[31] incorporate federated learning but rely on CNN-only or LSTM-based architectures respectively, with no Transformer integration and no explainability mechanism. Critically, no prior study combines all five dimensions simultaneously. The proposed framework is the first to integrate CNN-Transformer architecture, federated learning with FedAvg, LIME-based explainability, multi-class classification, and evaluation on the IoT-23 dataset within a single cohesive framework.

As evidenced by the comparative analysis in Tables 1 and 2, despite significant progress in applying deep learning and federated learning to IoT intrusion detection, four critical gaps remain unaddressed in the literature. First, most existing hybrid models, including CNN-Transformer architectures such as Ullah et al. and Yao et al. have not been integrated with federated learning, leaving data privacy a persistent concern in distributed IoT deployments. Second, model transparency is largely overlooked; very few works incorporate XAI techniques such as LIME within FL-based IDS frameworks, limiting the trustworthiness of automated detection decisions for security analysts. Third, many solutions focus exclusively on binary classification and show significantly degraded performance under multi-class attack scenarios, as demonstrated by Roy[18], who reported only 72% multi-class accuracy on the IoT-23 dataset. Fourth, generalizability is rarely validated across multiple independent datasets, raising questions about real-world applicability beyond the training distribution. The proposed framework directly addresses all four gaps by integrating a dual-block CNN-Transformer architecture with federated learning using FedAvg, LIME-based explainability, and comprehensive multi-class classification, validated on both the IoT-23 and CIC IoT-DIAD 2024 datasets.

To address these gaps, this study makes four distinct contributions not previously integrated in the IoT intrusion detection literature. First, a novel dual-block CNN-Transformer architecture is proposed that simultaneously captures local spatial features via convolutional layers and long-range temporal dependencies via multi-head self-attention, a combination not previously evaluated on IoT-23 under both binary and multi-class classification within a federated learning context. Second, the proposed hybrid model is integrated within a four-client federated learning framework using FedAvg aggregation, enabling privacy-preserving decentralised training without exposing raw IoT traffic data. Third, LIME-based explainability is incorporated to generate per-prediction feature attributions, enabling network security analysts to identify which traffic features drive attack classification decisions, representing the first integration of this XAI-FL combination evaluated on the IoT-23 dataset. Fourth, model generalizability is validated through a comprehensive ablation study on the CIC IoT-DIAD 2024 dataset, confirming robust performance across diverse IoT attack scenarios beyond the primary training distribution.

The power of CNNs lies in their ability to identify spatial patterns, making the detection of malicious IoT traffic more accurate. The self-aware mechanisms in Transformer models detect long-term relationships and temporal correlations in ordered data, improving our understanding of how attack behaviors change over time. When we combine feature extraction in CNNs with chronological modeling in Transformers, we make detection even more precise, especially when multiple IoT attacks occur simultaneously.

Methodology

This section describes the details regarding the datasets utilized and the experimental environment for all conducted experiments using the IoT-23 dataset. Following this, the process involves dataset preprocessing, feature selection, and the division of the dataset into training, validation, and testing subsets. Subsequently, various machine learning models are trained, including ensemble models, a CNN model, and a newly proposed explainable hybrid CNN transformer-based model that combines CNN and transformer architectures with a non-federated learning scenario in the first experiments. Additionally, a federated learning approach is applied to improve the performance of the proposed model and keep the privacy of data by training it with four different clients, as shown in Fig. 1. Finally, evaluation measures are used to gauge the effectiveness of the models that have been introduced.

Reference & Year	Dataset used	Methodologies	Results	Research gap
[10]Ullah et al. (2023)	Combined the three datasets of anomaly detection	CNN, Transformer	89.6% on the CUHK dataset	No federated learning integration; focused solely on anomaly detection without multi-class IoT attack classification or XAI explainability
[13]Ali et al. (2022)	DDoS evaluation dataset	CNN-LSTM	The performance ratio is 90.26%.	Limited to binary classification only; no privacy-preserving FL mechanism and no model interpretability component
[14]Umair et al. (2023)	N-BaIoT dataset	Spiking Neural Networks (SNNs)	71% of accuracy.	Low accuracy (71%); no FL integration and no XAI; limited generalizability across diverse IoT attack scenarios
[16]Booij et al. (2022)	ToN_IoT dataset, and IoT-23	RF, Gradient Boosting, and NN	97.84% for ToN_IoT, and 99.42% for IoT-23	Binary classification only using traditional ML; no deep learning hybrid architecture, no FL, and no interpretability
[17]Douiba et al. (2023)	IoT-23	RF, AdaBoost, and Catboost	99.9%for Catboost	Artificially balanced dataset limits real-world applicability; binary classification only with no FL or XAI component
[18]Roy et al. (2018)	IoT-23	Bi-directional LSTM	95% (single-class), 72% (multi-class)	Multi-class accuracy drops significantly (72%); no FL for privacy preservation and no explainability mechanism
[19]Liang & Vankayalapati (2021)	IoT-23	CNN, SVM, Decision Trees	73%	No comparative analysis; restricted to binary tasks with no FL or XAI, limiting deployment trustworthiness
[20]Alhamad & Alserhani (2022)	IoT-23	RF, CNN	89.22%	No hybrid architecture explored; no FL for decentralized privacy, no multi-class evaluation, and no interpretability
[21]Garg et al. (2022)	ableIoT-23	XGB, LGB	94.79%	Traditional boosting only; no deep learning hybrid, no FL integration, and no XAI for transparent decision-making
[22]Nazir et al. (2024)	IoT-23, N-BaIoT, CICIDS2017	CNN-LSTM	95%	CNN-LSTM evaluated across datasets but without FL for privacy preservation or XAI for interpretability
[31]Sorour et al. (2025)	Multiple datasets	LSTM-JSO	Accuracy from 0.923 to 0.940	Hybrid CNN-Transformer without FL integration or XAI; no privacy-preserving federated training evaluated

Table 1. Comparative analysis of related IoT intrusion detection studies, highlighting methodological approaches, performance results, and identified research gaps addressed by the proposed framework.

Reference	CNN-Transformer	Federated learning	XAI/LIME	Multi-class
Ullah et al. (2023)[10]	✓	✖	✖	✖
Yao et al. (2023)[41]	✓	✖	✖	✖
Roy (2018)[18]	✖	✖	✖	✓
Alhamad & Alserhani (2022)[20]	✖	✖	✖	✖
Torre et al. (2025)[34]	✖	✓	✖	✖
Sorour et al. (2025)[31]	✖	✓	✖	✓
Proposed framework	✓	✓	✓	✓

Table 2. Comparison of the proposed framework with closely related studies highlighting the unique combination of contributions.

Dataset description

To establish the validity of the proposed Transformer-CNN model, we used a dataset that is one of the most popular datasets for IoT attack detection and contains 23 different IoT devices. The dataset contains 21 benign and 23 malicious device classes, originating from multiple sources. Data was collected over 4 months and contains benign and attack traffic, such as DDoS, port scanning, spam, etc. The dataset contains over 20,000 attack samples, over 2 million benign samples, and over 320 PCAP files[44]. The dataset comes with 21 features, including labels grouped into 12 distinct categories. In this paper, we use most of the IoT-23 dataset except for four attacks out of twelve because of the lack of rows, including C&C-Mirai, C&C-HeartBeat-FileDownload, FileDownload, and Okiru, for both experiments, without federated and with federated learning. The IoT-23 dataset is unique in that it is specifically tailored for IoT environments, capturing realistic traffic from various smart devices infected with malware such as Mirai and Gafgyt, as well as benign IoT traffic.neering.

Dataset preprocessing

The processing of IoT-23 data unfolds in a structured, three-step manner. First, the task involves converting Packet Capture (PCAP) files into CSV format, allowing for easier handling and analysis. The combination of the pyshark, pandas, and openpyxl libraries were used to read PCAP files and export the data to excel format[45]. Subsequently, the focus shifts to addressing any missing values within the dataset, alongside the transformation of categorical data into numerical representations using the OrdinalEncoder library[46]. Upon obtaining the PCAP files, various tools are employed to serve as metadata and features, facilitating the extraction of essential metadata features from the data using the MinMaxScaler method[47].

Feature selection

Selecting relevant features from large datasets is a key step in reducing machine learning processing time and cost[48]. IoT-generated data is huge and complex and needs to be processed quickly. It also contains redundant and less reliable features[49].

Feature selection from the available IoT-23 dataset is most important to improve the accuracy and efficiency of future machine learning algorithms in classifying whether traffic is from a botnet[50]. Our proposed method uses a Random Forest (RF) model and removes less important features from the dataset, resulting in better performance of future machine learning algorithms[51]. In Fig. 2, a collection of notable features is displayed, each assessed against specific levels of significance. Any features that fell below the established threshold of 0.050 were excluded from consideration.

The RF method is distinguished by its minimal risk of overfitting, coupled with its capacity to offer a substantial degree of interpretability. As a result, 13 features were retained for model training, as listed in Table 3 below. These features capture key aspects of network traffic behavior, including flow duration, packet counts, byte volumes, connection states, and protocol types, which are the most discriminative attributes for distinguishing benign from malicious IoT traffic.

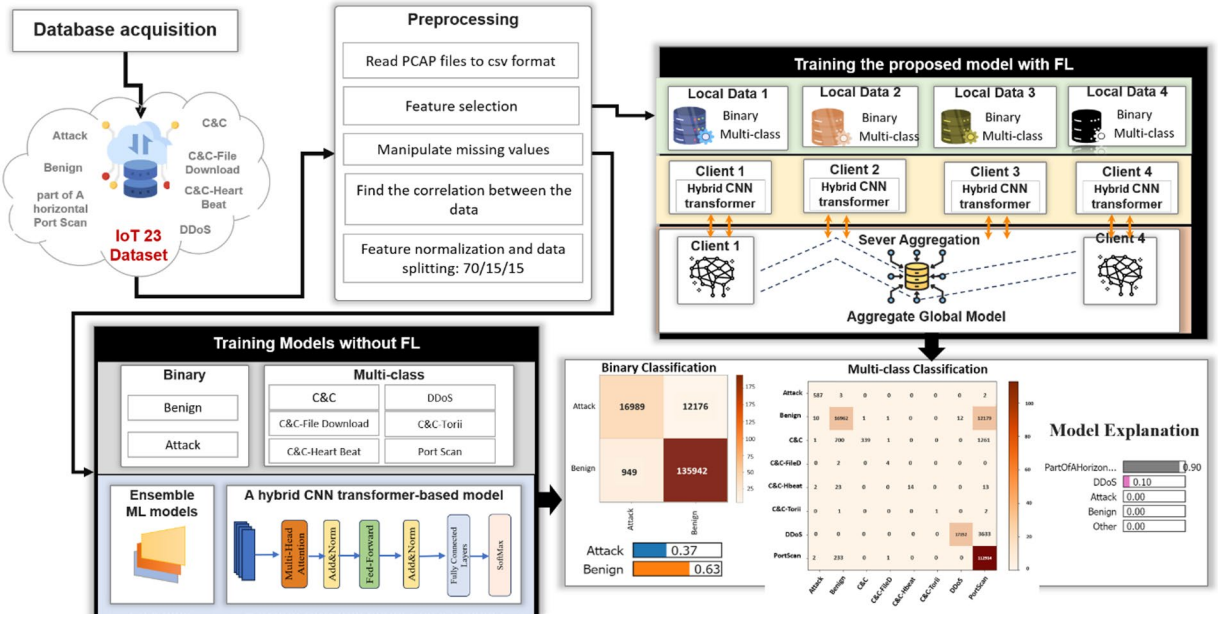


Fig. 1. The abstract of the proposed hybrid CNN transformer-based framework to predict IoT attacks. This model operates under both federated and non-federated learning scenarios, highlighting its versatility in different learning environments.

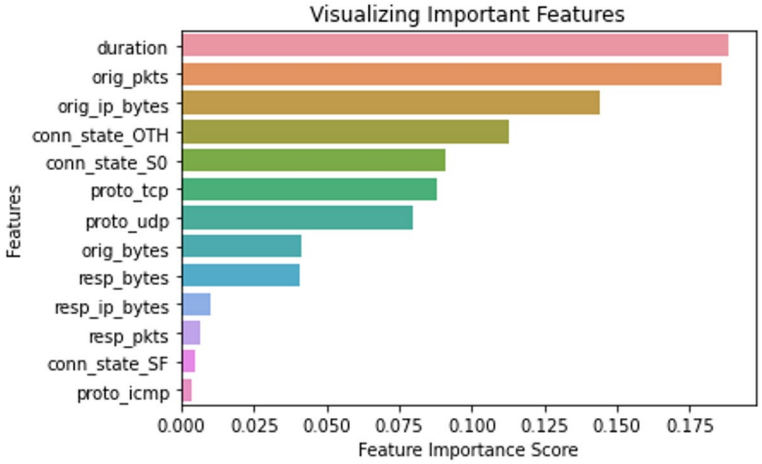


Fig. 2. The highest feature significance values were achieved through the use of a random forest technique.

#	Feature name	Importance score	Description
1	duration	0.185	Duration of network connection (seconds)
2	orig_pkts	0.183	Number of packets sent by originator
3	orig_ip_bytes	0.152	Total IP-level bytes from originator
4	conn_state_OTH	0.103	Connection state: other/ undefined state
5	conn_state_S0	0.096	Connection state: SYN sent, no reply (half-open)
6	proto_tcp	0.091	Protocol type: TCP (binary encoded)
7	proto_udp	0.083	Protocol type: UDP (binary encoded)
8	orig_bytes	0.037	Payload bytes sent by originator
9	resp_bytes	0.033	Payload bytes sent by responder
10	resp_ip_bytes	0.019	Total IP-level bytes from responder
11	resp_pkts	0.014	Number of packets sent by responder
12	conn_state_SF	0.008	Connection state: normal established and closed
13	proto_icmp	0.006	Protocol type: ICMP (binary encoded)

Table 3. Final selected features from the IoT-23 dataset after Random Forest-based feature selection (threshold = 0.050).

Dataset splitting data for training, validation, and testing

Once the dataset has been meticulously prepared, it undergoes a random partitioning process based on class categories. This division allocates 70% of the data for training purposes, while 15% is reserved for validation, and the remaining 15% is designated for testing. This strategic division is consistently applied across both non-federated and federated learning scenarios.

The organization of the dataset post-splitting for class categories is succinctly outlined in Tables 4 and 5, which illustrate the distribution for both binary and multi-class approaches within the non-federated and federated learning scenarios. To address the challenge of imbalance, particularly within the training sets across all classification tasks, we explore the application of the Synthetic Minority Oversampling Technique (SMOTE). This method is designed to enhance balance by augmenting the dataset focused on the majority class. However, despite these efforts, there was no measurable improvement in accuracy, mainly due to the substantial disparity between the majority class, which comprises 753,565 instances, and the minority class, which consists of only 30 instances.

Approach	Label with (Encoding number)	Total instances	A training set 70%	A validation set 15%	A testing set 15%
Binary	Normal (0)	195,270	137,037	29,068	29,165
	Attack (1)	911,771	637,891	136,989	136,891
Multi-class	PartOfPortScan (7)	753,565	527,389	113,026	113,150
	Benign (1)	195,270	137,037	29,068	29,165
	DDoS (6)	138,777	96,914	21,078	20,785
	C&C (2)	15,100	10,568	2,230	2,302
	Attack (0)	3,915	2,719	604	592
	C&C-HeartBeat (4)	341	243	46	52
	C&C-FileDownload (3)	43	35	2	6
	C&C-Torii (5)	30	23	3	4

Table 4. Data partitioning for training, validation, and testing in non-federated scenarios for both binary classification and multi-class classification approaches.

Approach	Label with (Encoding number)	Total instances	Client1	Client2	Client3	Client4
Binary	Normal (0)	195,270	48,767	48,507	49,083	48,913
	Attack (1)	911,771	227,994	228,253	227,677	227,847
Multi class	PartOfAHorizontalPortScan (7)	753,565	188,367	188,556	188,295	188,347
	Benign (1)	195,270	48,825	48,663	48,643	49,139
	DDoS (6)	138,777	34,649	34,718	34,958	34,452
	C&C (2)	15,100	3,862	3,733	3,749	3,756
	Attack (0)	3,915	953	990	1,011	961
	C&C-HeartBeat (4)	341	92	77	86	86
	C&C-FileDownload (3)	43	9	15	10	10
	C&C-Torii (5)	30	4	8	8	9
	Total		276,763			

Table 5. Data partitioning for the purposes of training, validation, and testing in federated scenarios for both binary classification and multi-class classification approaches.

Tables 3 and 4 present the distribution of attack categories in the IoT-23 dataset used for both binary and multi-class classification experiments. Each attack type is assigned a unique integer label for model training: Benign traffic is encoded as class 1, while each attack category receives a distinct integer identifier (e.g., C&C = 2, DDoS = 3, PartOfAHorizontalPortScan = 4, and so forth), as shown in the "Label with Encoding number" column. This encoding scheme enables the model to perform both binary classification (Benign vs. Attack) and multi-class classification across all eight retained attack categories simultaneously.

Non-federated learning scenario

Non-federated, or centralized, experiments in machine learning rely on one centralized server for data collection, storage, and processing. Data is stored and processed on a single machine, making non-federated methods inherently centralized. This allows easy access to stable, private, and homogenously distributed datasets[52]. In this experiment, the dataset was used exclusively on one server for all training purposes. We used recent machine learning classification techniques, such as ensemble models, CNN, and compared the results with the proposed model. The hybrid CNN transformer-based model combines the advantages of both CNN and transformer-based techniques. The following sections provide a detailed description of each technique.

Machine learning models and CNN

In the context of non-federated learning, we utilized XGBoost[53], Catboost[54], Random Forest[55], Voting Ensemble[56], and a CNN model. All machine learning models were fine-tuned with n_estimators set to 160 and random_state to 14. Additionally, the ensemble model was created using a soft voting classifier, which combined the top three accuracy models: XGBoost, Catboost, and RF. The hybrid CNN transformer-based model produced improved results and outperformed the previous models in terms of accuracy.

CNNs are deep learning algorithms commonly used in cybersecurity. They are effective for feature extraction in visual data, making them valuable in various security systems[57]. CNNs have been applied in malware detection, deception honeypots, and network intrusion detection. Intrusion detection is a rapidly developing field for CNN applications. By identifying abnormal network traffic, CNNs can provide accurate results while

minimizing computational costs and bandwidth usage[58]. CNNs can offer better feature extraction capabilities than traditional classification methods.

CNN-transformer-based model

Transformers and CNNs are widely used architectures in the field of NLP and image processing, respectively. Looking at them individually in their respective domains, the Transformer model is cumbersome in terms of time, memory, and operations that need to be performed[59]. The attention mechanism used in the fully connected layers of the transformer should be irrelevant at the word level and should just evaluate the relationship through the features. This makes the transformer model difficult to explain at the word level[60]. In addition, CNN models cannot learn long patterns and only extract local features when the kernels stay small, with a maximum sentence length[61]. Based on the analysis, the rich semantic contextual information of the document seems to be the basis of the successful operation of the end-to-end learning model in NLP tasks. The self-attention mechanism can generate a representation of a single input sequence by establishing connections between different positions within the sequence[62,63]. Meanwhile, computed attention functions are stored in the attention matrix, which also includes the set of queries (Q), keys (K), and values (V) as computed in this equation:

$$Attention\,(Q, K, V) = softmax\left(\frac{Qk^2}{\sqrt{d_k}}\right) v \tag{1}$$

Given an input with a dimension d_k for queries and keys and a dimension d_v, the dot product of all queries with keys is calculated by normalizing each by $\sqrt{d_k}$ and applying SoftMax to determine the weights for the value pairs.

Multi-headed attention enables the model to concurrently focus on inputs from many representation subspaces across different locations.

$$MultiHead\,(Q, K, V) = Concat\,(head1, \ldots\ldots, headh)\,W \tag{2}$$

In this paper, we propose a CNN transformer-based model as an IoT attack detection solution. This transformer model is specifically designed for sequence-to-sequence tasks and draws inspiration from transformer network architecture, as illustrated in Fig. 3. It leverages Keras and TensorFlow's layers to construct an encoder-decoder framework with multi-head self-attention mechanisms.

The model hyperparameters include a vocabulary size (input_vocab_size) of 1024 for the embedding layer, which defines the size of the discrete input space after feature quantization of the network flow features into 1024 unique integer indices. This discretization step maps continuous network flow features (such as protocol type, connection state, and traffic statistics) into integer tokens, enabling the Transformer's embedding layer to process them. The model dimensionality (embedding and transformer layer size) is 256, with 8 attention heads, a feed-forward layer dimension of 256, a dropout rate of 0.1 for regularization, a maximum sequence length (maxlen) of 50, and an early stopping patience of 11 during training.

The proposed model has 2 convolutional blocks that use the TransformerBlock_Encode and TransformerBlock_Decode layers. Each block has 256 filters, a kernel size of 5, and a stride of 1. The CNN model does not have any pooling layers because the attention mechanism has taken the place of pooling. After each attention and feed-forward layer, layer normalization is used with a dropout rate of 0.1. The Adam optimizer is used with a learning rate of 0.001, a batch size of 32, and for 50 epochs. The input layer has an embedding layer with input_vocab_size = 1024, a Transformer Block Encode with a feed-forward layer with ff_dim = 256, a multi-head attention layer, and a feed-forward layer with ff_dim of 256. The Transformer Block Decoder has a multi-head attention layer with eight heads. The fully connected layers consist of dense layers with 512, 256, 128, and 8 units for multi-class classification. By applying the regularization techniques (dropout, early stopping, feature selection, cross-validation, and separate validation sets). In addition, federated learning trains models on diverse client data, and XAI verifies the model's decisions. The proposed models mitigate the overfitting.

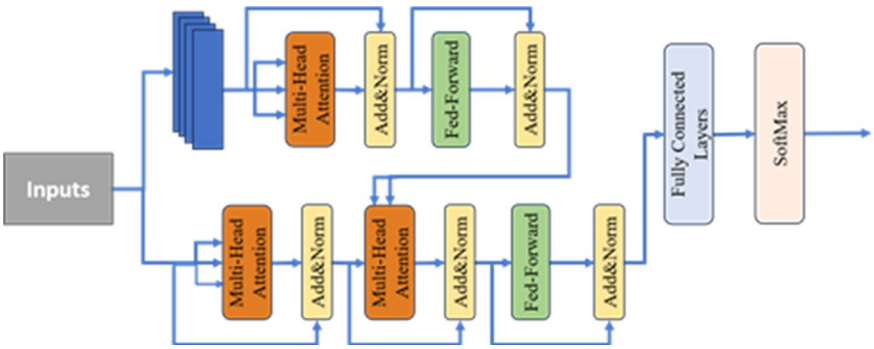


Fig. 3. The proposed hybrid CNN transformer-based model that applied for non-federated and federated learning scenarios.

Federated learning scenario

Federated learning is a concept that does not depend on the central collection of training data[64]. It is an extension of distributed learning or collaborative learning, yet it focuses on distributing the model training process to multiple nodes without the need to collect the data from one of the nodes[65]. For instance, federated learning can help reduce the power consumed by an IoT device if its limited data is used repeatedly in transmission[66]. Additional example can be found in a medical-related application, where the local node whose data is used cannot transfer its data outside of the hospital for the patient's privacy protection, but it can use the other hospital's data through the concept of federated learning[67].

In the context of this study, FL can provide a reasonable solution to the security of IoT devices. In our current implementation, the client data split is assumed to be Independent and Identically Distributed (IID) for simplicity, given that the IoT-23 dataset used in our study contains data from various devices that share a similar distribution of data across all clients. However, we acknowledge that this is a simplifying assumption that may not reflect real-world IoT deployments, where data distributions across devices are typically heterogeneous and non-IID due to differences in device types, locations, and traffic patterns. Under non-IID conditions, the FedAvg algorithm is known to suffer from client drift, where local models diverge significantly from the global optimum, potentially degrading global model accuracy by 5–15% compared to IID settings, as reported in the literature. To mitigate this in future work, personalised federated learning strategies such as FedProx, which introduces a proximal term to limit local model divergence, or per-FedAvg, which incorporates model-agnostic meta-learning, could be adopted to improve convergence under non-IID data distributions. However, we acknowledge that this is a simplifying assumption that may not reflect real-world IoT deployments, where data distributions across devices are typically heterogeneous and non-IID due to differences in device types, locations, and traffic patterns. Under non-IID conditions, the FedAvg algorithm is known to suffer from client drift, where local models diverge significantly from the global optimum, potentially degrading global model accuracy by 5–15% compared to IID settings, as reported in the literature. To mitigate this in future work, personalized federated learning strategies such as FedProx, which introduces a proximal term to limit local model divergence, or per-FedAvg, which incorporates model-agnostic meta-learning, could be adopted to improve convergence under non-IID data distributions. This paper develops and validates an explainable CNN transformer-based model as a new attack detection model, and the main contribution is based on the transformer and applies it to the federated learning scheme. However, this doesn't mean the paper suggests federated learning for the first time. The precision of the CNN model is improved by using transformer techniques and passing it through four clients, which aggregate the results of all local models into the global model. The data is divided for each client, with the models being trained separately for 50 epochs before the weights are updated between the local models and the server over 10 rounds. The steps for the federated learning process are shown in Algorithm 1. The details of the model's hyperparameters in a federated learning setting are provided in Table 6.

Variables	Description
Number of Clients = 4	Number of participants
Data splitting per client	Splitting data for training, validation, and testing per client, 80%,10%, and 10%, respectively
Number of labels 2 and 8	Eight types of IoT attacks
learning_rate = 0.001	Learning rate
Batch_size = 32	Batch size
Epochs = 50	Train a local model per client
Early stopping patience	10
Number of rounds = 10	Number of rounds to update weights
Aggregation method	FedAvg tensorflow_federated.learning.build_federated_averaging_process(createModel, client_optimizer_fn)

Table 6. The parameters of the CNN transformer-based model are used to update weights between clients and server aggregation.

Start:

Initialize the global model parameters θ

Declare a list of clients C_i

For each round r do

  for each client k in C_i

    Training a local model of client k by the ET-CNNFed model

    Sending the local model of k to Server S

    Retrieve global model parameters θ from Server S

    Update the local model of k

    end for

  Compute global model update $A\theta = Average\ (A\theta i\ for\ each\ k\ in\ Ci)$

  Update the global model θ

end for

Return global model parameters θ

End

Algorithm 1: The proposed hybrid explainable CNN transformer-based model in the federated learning scenario.

For the local training on each client, each client k, the client has a local dataset D_k. In each round, the client performs local gradient descent to update the local model wFL_k. The update is done as follows:

$$\mathrm{w}FL_k^{t+1} = \mathrm{w}FL_k^t - n\nabla L\left(\mathrm{w}FL_k^t,\ D_k\right) \tag{3}$$

Where $\mathrm{w}FL_k^{t+1}$ is the model weights for client k at round t, n is the learning rate, $L\left(\mathrm{w}FL_k^t,\ D_k\right)$ is the loss function evaluated on the client D_k. and ∇ is the gradient of the loss function[68].

Explanation AI model

Explainable Artificial Intelligence (XAI) has emerged as a response to the black-box nature of many machine learning models. XAI seeks to add a layer of transparency to systems' decision-making processes, aiming to capture the reasoning that is used by humans[69]. LIME is crucial in enhancing the transparency and interpretability of our hybrid CNN-Transformer model, especially in cybersecurity contexts where trust and transparency are paramount. We have now clarified the steps involved in applying LIME, such as how it generates local surrogate models to explain the predictions of our complex model, thereby helping users understand why certain features lead to specific attack classifications[70]. We highlight how the feature importance scores derived from LIME are utilized to identify the most influential features for each prediction. For example, LIME might show that specific features like proto_udp or orig_ip_bytes are strong indicators of benign traffic, while features like attack_type contribute significantly to detecting DDoS or C&C attacks, as presented in the mathematical equation:

$$min_{g\in G} = \sum_i \boldsymbol{\pi}\ (x_i,\ \mathrm{g})\ .\ \ L\ (f, g) +\ \boldsymbol{\Omega}\,(\mathrm{g})\,, \tag{4}$$

Where π (x_i, g) represents the proximity function, measuring how close the perturbed sample x_i is to the original instance. L(f, g) is the loss function, which measures how well the surrogate model g approximates the behavior of the complex model f in the local neighborhood of x_i. Ω(g) is a regularization term that controls the complexity of the surrogate model[71].

Feature contribution values will vary between − 1 and 1. When the values are positive, they indicate the abnormal class, while negative values indicate the normal class. The Lime library, specifically LimeTabularExplainer, applies LIME explanations using parameters such as training, classes, and kernel[72].

Evaluation

The models were evaluated again using the test data and taking into account specific detection criteria as used in our previous work[73]. Accuracy is a measure of the percentage of all accurate predictions. It demonstrates the overall performance of the model for both positive and negative classes, and is especially beneficial when the classes are evenly distributed. It is determined by the following formula,

$$Accurac\ (ACC) = \frac{True\ Positive\ (TP) + True\ Negative\ (TN)}{True\ Positive\ (TP) + True\ Negative\ (TN) + False\ Positive\ (FP) + False\ Negative\ (FN)} \quad (5)$$

Precision evaluates the accuracy of positive predictions by measuring the proportion of correctly predicted positive results out of all instances predicted as positive. This metric is particularly beneficial for minimizing the occurrence of false positives, as it emphasizes the correctness of positive predictions rather than capturing all actual positive outcomes. It can be calculated as follows,

$$Precision\ (PRE) = \frac{True\ Positive\ (TP)}{True\ Positive\ (TP) + False\ Positive\ (FP)} \quad (6)$$

Recall, also referred to as sensitivity or the True Positive Rate, calculates the percentage of actual positive results that the model accurately identifies. Mathematically, it is expressed as,

$$Recal\ (REC) = \frac{True\ Positive\ (TP)}{True\ Positive\ (TP) + False\ Negative\ (FN)} \quad (7)$$

The F1-score is a metric that measures the ratio between Precision and Recall by calculating their Harmonic Mean.

$$F1 - Score = \frac{2Precision \times Recall}{Precision + Recall} \quad (8)$$

The Area Under the Curve (AUC) is a metric that evaluates a classifier's ability to differentiate between classes using the ROC curve. A high AUC suggests that the model can accurately differentiate between positive and negative samples. The ROC curve illustrates the balance between false positives and true positives, offering important insights for assessing the classifier. We can measure the occurrence of false positives by comparing the misclassified negative samples with the total number of negative samples.

The 95% Confidence Interval (CI)

A 95% CI is a range of values, calculated from sample data, that likely contains the true population mean, meaning if you repeated the study many times, 95% of the intervals calculated would capture the true value, using a formula:

$$CI = Mean\ \pm\ Z * \frac{SD}{\sqrt{n}} \quad (9)$$

Where the mean is the sample mean (average) of the data. Z is the Z-score corresponding to the desired confidence level. For a 95% confidence level, the Z-score is 1.96. Standard Deviation (STD) is the standard deviation of the sample data, and n is the sample size.

$$STD = \sum_{i=1}^{n} \frac{1}{n} * (ACC_i - Mean) \quad (10)$$

Execution environment

The study used an MSI GS66 laptop containing an Intel Core i7 11th generation (11800 H) processor, 32 GB of RAM, a 2 TB SSD NVMe storage, and an RTX 3080 graphics card with 16 GB of memory. The described trials were performed on a Windows 11 operating system using Python 3.10 as well as the Keras and TensorFlow backend libraries.

Results

In this section, we presented and thoroughly examined the outcomes of the IoT attack detection utilizing the IoT-23 dataset in two distinct scenarios: one involving non-federated learning and the other involving federated learning. We delved into each scenario, exploring both binary classifications and multi-class classification approaches. These approaches were implemented using machine learning ensemble models, CNN, and the groundbreaking hybrid CNN transformer-based model that we propose. The CNN model and the suggested model have been tested once more in the federated scenario, demonstrating the durability of anomaly detection for IoT networks in a federated learning scenario. The results demonstrated that the hybrid CNN transformer-based model was capable of efficiently minimizing the detection error along with the traffic, with all sorts of attacks. The robustness of the model was also evaluated with the detection of the proposed attack present in the federated learning scenario, which makes it well-suited to be utilized to keep the privacy of IoT devices.

Non-federated experiments scenario

In the non-federated experiment scenario, we employ both binary classification and multi-class classification approaches that include a total of eight types of IoT attacks: PortScan, benign, DDoS, C&C, Attack, C&C-HeartBeat, C&C-FileDownload, and C&C-Torii. These attacks provide a comprehensive and diverse range of scenarios to analyze and predict. Utilizing advanced AI ensemble models, namely XGBoost, Catboost, Random Forest, and Voting Ensemble, allows us to maximize the accuracy and effectiveness of our predictions. Furthermore, to further enhance the prediction capabilities, we also employ a CNN model. Additionally, a

hybrid CNN transformer-based model is utilized, bringing together the strengths of both CNN and transformer architectures. This holistic approach ensures accurate and robust predictions for detecting and mitigating IoT attacks, thereby enhancing the security and resilience of IoT systems. The combined voting model incorporates three different machine learning models using a soft voting approach. The proposed hybrid CNN transformer-based model creates an encoder-decoder framework with multi-head self-attention mechanisms. The evaluation begins with the performance of the ensemble machine learning models, followed by the evaluation of the CNN model and the enhancements provided by the proposed hybrid CNN transformer-based model.

Binary classification approach

The evaluation of IoT attack detection results for all trained models using the binary approach is shown in Table 7. The analysis presents comprehensive insights into the effectiveness and accuracy of the models in detecting IoT attacks. The table provides a detailed overview of the performance metrics, such as accuracy, precision, recall, and F1 score, for each model, allowing for a comprehensive comparison. In the binary approach, the ensemble ML models demonstrated exceptional classification accuracy. Specifically, the models achieved accuracy rates of 91.69%, 91.71%, and 91.80% for RF, XGboost, and Catboost, respectively. Notably, when the previous three models were merged, the ensemble voting model exhibited a slight improvement in accuracy, reaching a rate of 91.82% and an F1 score of 90.94%. Furthermore, it obtained an impressive AUC of 77.83%. While the AUC may appear slightly lower compared to the accuracy metric, it is important to consider the unbalanced nature of all eight labels. Despite this challenge, the achieved F1 score stands out as one of the best results when compared to previous studies. These findings highlight the efficacy of the binary approach in utilizing ensemble ML models to achieve remarkable accuracy in classification tasks.

After that, the CNN model is used to compare its performance by itself and with the enhancement using a transformer and self-attention. The CNN model showed lower effectiveness compared to the combined ML models, reaching an accuracy rate of 87.61% and 89.23% precision. The proposed hybrid CNN transformer-based model surpasses the combined ML models and the CNN model. It attains an accuracy of 92.10% and an F1 score of 91.31%. With the inclusion of the transformer and self-attention mechanisms, the performance of the CNN model improves significantly. The accuracy rate increases by 4.49% points, reaching a remarkable 92.10%. Moreover, the precision also sees a notable improvement, with a precision rate of 89.23%, surpassing the initial result of the CNN model alone. Additionally, the hybrid CNN transformer-based model achieves an impressive F1 score of 91.31%, indicating its effectiveness in accurately classifying the data. The incorporation of the transformer and self-attention components proves to be a valuable enhancement to the CNN model, surpassing the performance of the ensemble ML models as well.

In Fig. 4, displayed the rates of true positive, true negative, false positive, and false negative for the CNN model, ensemble voting model, and hybrid CNN transformer-based model in the confusion matrix. The CNN model incorrectly predicted just one traffic instance as an attack in the benign class, but it mispredicted approximately twenty thousand traffic instances as an attack for the benign class, with only 8587 being true positives out of a total of 29,165 benign cases. The ensemble model was made to increase the TP rate predictions to 16,412, more than double the initial count. In addition, the proposed model enhances the TP rate to 16,700, showing a significant improvement in prediction accuracy. Both the ensemble model and the proposed hybrid model have greatly increased the number of true positive predictions, making them more reliable and effective in identifying attacks correctly.

Multi-class classification approach

For the multi-class classification, the evaluation of IoT eight attack detection results for all trained models is shown in Table 8. The ensemble machine learning models showed a slight decrease in accuracy compared to binary classification for eight attack types. Despite this, the results are still considered impressive when compared to previous research on multi-classification. The ensemble ML models had accuracy percentages of 88.61%, 88.62%, and 88.71% for RF, XGboost, and Catboost. It is worth noting that when these three models were combined, the ensemble voting model showed a small increase in accuracy, reaching 88.73% and an F1 score of 87.45%. Additionally, it achieved an impressive AUC of 82.82%, which is slightly superior to the binary approach metric despite the unbalanced nature of all eight labels.

Models	ACC	REC	PRE	F1-score	AUC
XGboost	91.71	91.71	91.91	90.84	77.78
RF	91.69	91.69	92.01	90.76	77.36
Catboost	91.80	91.80	92.05	90.92	77.83
Ensemble voting	91.82	91.82	92.09	90.94	77.83
CNN	87.61	87.61	89.23	84.66	64.72
Proposed CNN transformer-based model without FL	92.10	92.10	92.29	91.31	78.78

Table 7. The experimental evaluation of the ensemble machine learning and CNN models is contrasted with the proposed hybrid explainable CNN transformer-based model in a non-federated scenario using a binary approach.

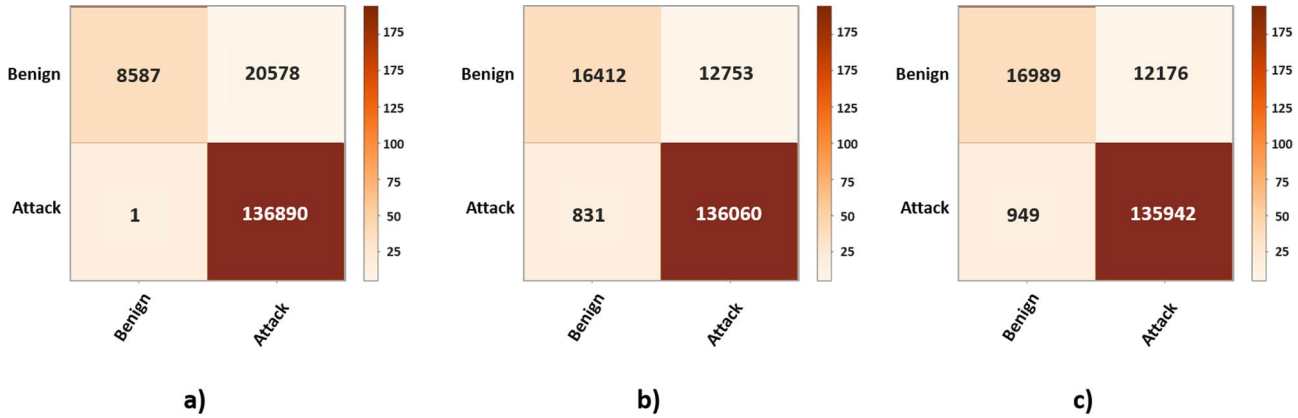


Fig. 4. The confusion matrices of the proposed models in non-federated scenarios are (a) the CNN model, (b) the ensemble model, and (c) the proposed hybrid CNN transformer-based model using a binary classification approach.

Models	ACC	REC	PRE	F1-score	AUC
XGboost	88.62	88.62	89.65	87.35	82.76
RF	88.61	88.61	89.27	87.36	82.73
Catboost	88.71	88.71	89.81	87.44	82.81
Ensemble voting	88.73	88.73	89.84	87.45	82.82
CNN	83.73	83.73	85.58	80.47	75.10
Proposed CNN transformer-based model without FL	89.11	89.11	90.07	87.97	83.49

Table 8. The experimental evaluation of the ensemble machine learning and CNN models is contrasted with the proposed hybrid explainable CNN transformer-based model in a non-federated scenario using a multi-class approach.

The following step involves using the CNN model to evaluate its performance both on its own and with the addition of a transformer and self-attention. The CNN model displayed reduced effectiveness compared to the combined ML models, achieving an accuracy rate of 83.73% and a precision of 85.58%. In addition, the results demonstrate that the incorporation of a transformer and self-attention greatly improves the performance of the CNN model. The hybrid model not only outperforms the combined ML models but also exceeds the capabilities of the standalone CNN model. By leveraging the power of the transformer architecture and self-attention mechanism, the accuracy rate of the hybrid model is boosted to an impressive 89.11%, showcasing its superiority in comparison to other models. Furthermore, the F1 score of the hybrid CNN transformer-based model stands at an exceptional 87.97%. This metric, which considers both precision and recall, further supports the effectiveness and robustness of the proposed model. The substantial improvement in both accuracy and F1 score highlights the significance of integrating a transformer and self-attention into the CNN architecture.

Almost all training models in our study have demonstrated varying degrees of incorrect predictions for these four classes, benign, C&C, DDoS, and port scan, as shown in Fig. 5. Surprisingly, CNN generated an astonishing 20,150 instances of mispredictions for the benign class while also incorrectly classifying all instances of C&C attacks. Furthermore, it wrongly labeled 3,784 instances of DDoS attacks and 298 instances of port scans. However, the ensemble ML model helped improve the prediction accuracy for C&C attacks. This enhanced model correctly predicted 241 instances of C&C attacks. The proposed CNN transformer-based model achieved remarkable results by correctly identifying 339 instances of C&C attacks, surpassing the performance of the previous models. Not only did our proposed model excel in C&C attack predictions, but it also significantly reduced the mispredictions for the benign class. Compared to the other models, our proposed model successfully reduced the mispredicted benign instances to 12,179 instances.

Table 9 presents the per-class breakdown of evaluation merices for each of the eight attack categories in the multi-class classification task, providing a finer-grained view of model performance that overall weighted averages alone cannot reveal. These per-class results are essential for understanding where the model excels and where performance is constrained by class imbalance, particularly for minority classes such as C&C with accuracy of 14.73% and C&C-HeartBeat with accuracy of 30.77%.

The per-class results in Table 9 reveal an important pattern that warrants careful interpretation. Per-class evaluation was conducted exclusively on the proposed model as it represents the primary contribution of this study. The C&C class ACC of 14.73% and F1 of 25.67% reflect an extreme dataset constraint, only 241 training

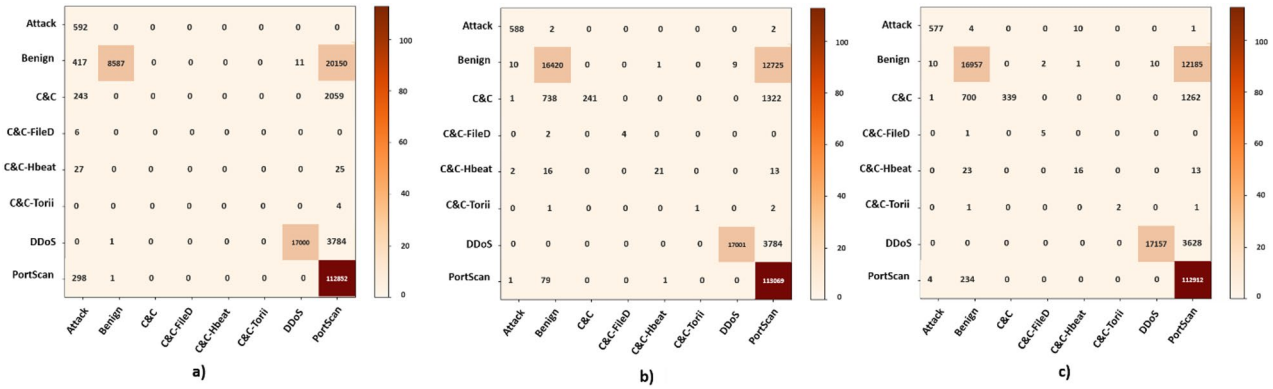


Fig. 5. The confusion matrices of the proposed models in non-federated scenarios are (a) the CNN model, (b) the ensemble model, and (c) the proposed hybrid CNN transformer-based model using a multi-class classification approach.

Class	ACC	REC	PRE	F1
PartOfAHorizontalPortScan (7)	99.79	99.79	86.85	92.87
Benign (1)	58.14	58.14	94.63	72.03
DDoS (6)	82.55	82.55	99.94	90.41
C&C (2)	14.73	14.73	1.00	25.67
Attack (0)	97.47	97.47	97.47	97.47
C&C-HeartBeat (4)	30.77	30.77	59.26	40.51
C&C-FileDownload (3)	83.33	83.33	71.43	76.92
C&C-Torii (5)	50.00	50.00	1.00	66.67
Average by 'weighted'	89.11	89.11	90.07	87.97

Table 9. The per-class performance result of proposed model for each of the attack types in the multi-class classification task.

Metric	Mean	STD	95% CI
ACC	64.60	29.32	[63.46, 65.74]
REC	64.60	29.32	[63.46, 65.70]
PRE	63.95	38.55	[62.76, 65.13]
F1	70.32	23.96	[69.20, 71.43]

Table 10. The computed statistical significance and 95% CI for the evaluation metrics in multiclass classification.

instances against 753,565 benign instances (ratio 1:3,125). Critically, 5 out of 8 attack classes achieve F1 > 76%, confirming that the model's weakness is class-specific and directly proportional to sample scarcity, not a systemic architectural failure. This is consistent with the literature, Roy et al.[18] reported 72% overall multi-class accuracy on IoT-23 under similar imbalance, while our weighted F1 of 87.97% represents a 15.97% point improvement.

The analysis presented in Table 10 focuses on the statistical tools that provides an estimate of the range in which to fall with 95% confidence. The mean, STD, and the 95% C in this analysis were calculated for the evaluation key metrics. These metrics are essential for understanding the stability and reliability of the model's performance.

The mean for both accuracy and recall are 64.60, indicating a relatively strong performance in these two metrics. In contrast, the F1 score, with a mean of 70.32, demonstrates a better balance between precision and recall, reflecting the model's ability to perform well on both fronts simultaneously. For the STD, both ACC and REC have a STD of 29.32, indicating moderate variability across different classes. On the other hand, PRE has a higher STD of 38.55, indicating more significant variation in precision across the classes. The F1 score, with a STD of 23.96, shows relatively more stability in the model's performance.

The 95% CI offer insights into the model's stability and reliability. For ACC and REC, the narrow CIs of [63.46, 65.74] and [63.46, 65.70] indicate high confidence in the model's consistent performance. In contrast, PRE has a wider CI of [62.76, 65.13], suggesting more variability in precision. The F1 score has the tightest CI of [69.20, 71.43], indicating consistent performance in balancing precision and recall across classes.

Approach	Models	1st round Accuracy	10th round Accuracy
Binary classification	CNN with FL	88.02	91.46
	Proposed CNN transformer-based model with FL	92.76	94.89
Multi-class classification	CNN with FL	85.00	88.26
	Proposed CNN transformer-based model with FL	89.43	92.17

Table 11. The experimental evaluation of the CNN model and the proposed hybrid explainable CNN transformer-based model through a federated learning scenario in binary and multi-class classification approaches.

Model	Task	Round 1 Accuracy (%)	Round 10 Accuracy (%)	Gain (pp)	Superiority over CNN at R10 (pp)
Proposed CNN-Transformer	Binary	92.76	94.89	+2.13	+3.43
CNN with FL	Binary	88.02	91.46	+3.44	—
Proposed CNN-Transformer	Multi-class	89.43	92.17	+2.74	+3.91
CNN with FL	Multi-class	85.00	88.26	+3.26	—

Table 12. Round-by-round convergence analysis of the proposed CNN-Transformer model versus CNN baseline under federated learning.

The next section will discuss the federated IDS experiments scenario. Federated learning, on the other hand, enables distributed training across edge devices, preserving data locality and privacy, which is crucial in sensitive or heterogeneous IoT networks.

Federated experiments scenario

The experimental evaluation in a federated learning (FL) scenario for both binary and multi-class classification was conducted using two models: a conventional CNN and a proposed hybrid explainable CNN transformer-based model. The performance was compared at the 1st and 10th rounds of training, and the results are presented in Table 11.

For binary classification, the CNN model with FL achieved an accuracy of 88.02% in the 1st round, which improved to 91.46% by the 10th round. In contrast, the proposed CNN transformer-based model with FL showed superior initial performance, achieving 92.76% accuracy in the 1st round and further improving to 94.89% by the 10th round. In multi-class classification, the CNN with FL reached 85.00% accuracy in the 1st round and improved to 88.26% by the 10th round. The proposed CNN transformer-based model with FL again outperformed the CNN, achieving 89.43% accuracy in the first round and 92.17% in the 10th round in multi-class classification.

Convergence analysis

Table 12 presents the round-by-round convergence behaviour of both models across binary and multi-class classification tasks. The proposed CNN-Transformer model demonstrates consistent convergence across all 10 federated communication rounds, achieving accuracy gains of 2.13 and 2.74% points in binary and multi-class tasks respectively. Notably, the proposed model achieves a higher Round 1 accuracy (92.76% binary, 89.43% multi-class) than the CNN baseline at Round 10 with 91.46% binary and 88.26% multi-class, indicating superior model initialisation stemming from the dual-block CNN-Transformer architecture. The performance gap between the two models remains stable across all rounds, 3.43% points in binary and 3.91 in multi-class at Round 10, confirming that the architectural advantage is consistent and not round-dependent.

The results indicate that the proposed hybrid CNN transformer-based model outperforms the standard CNN model in binary and multi-class classification tasks within the federated learning framework, as shown in Fig. 6. In the binary classification task, the CNN transformer-based model exhibited a higher initial accuracy of 92.76%, as compared to 88.02% for the standard CNN. By the 10th round of training, the gap between the two models increased, with the hybrid model achieving an accuracy of 94.89%, significantly outperforming the CNN, which achieved 91.46%. Similarly, the proposed model again showed superior performance in the multi-class classification task, starting at 89.43% accuracy in the 1st round compared to CNN's 85.00%. By the 10th round, the hybrid model reached 92.17%, while the CNN achieved 88.26%. This demonstrates the robustness and efficiency of the proposed model in handling more complex classification tasks within the federated learning environment.

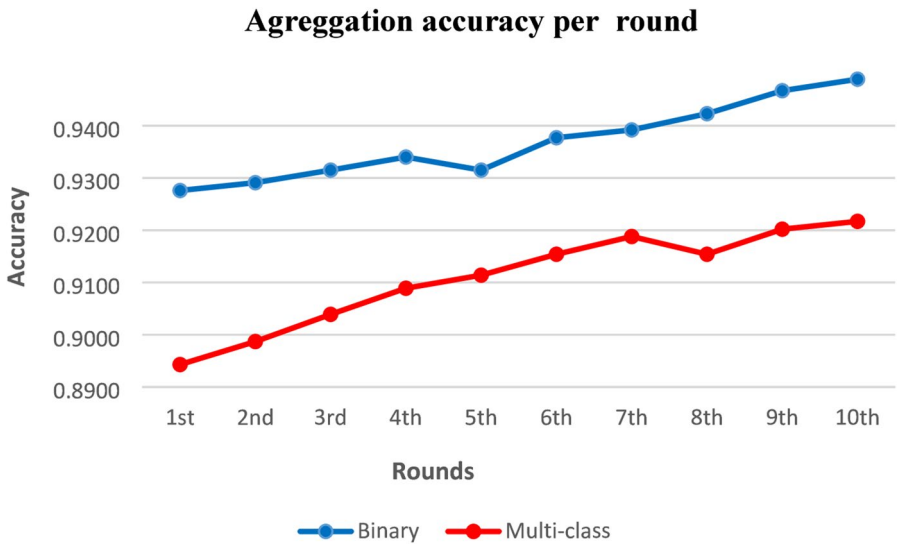


Fig. 6. The level of accuracy of both CNN and the CNN transformer-based model per round of updated weights to the server's global model in federated learning scenario for binary and multi-class classification approaches.

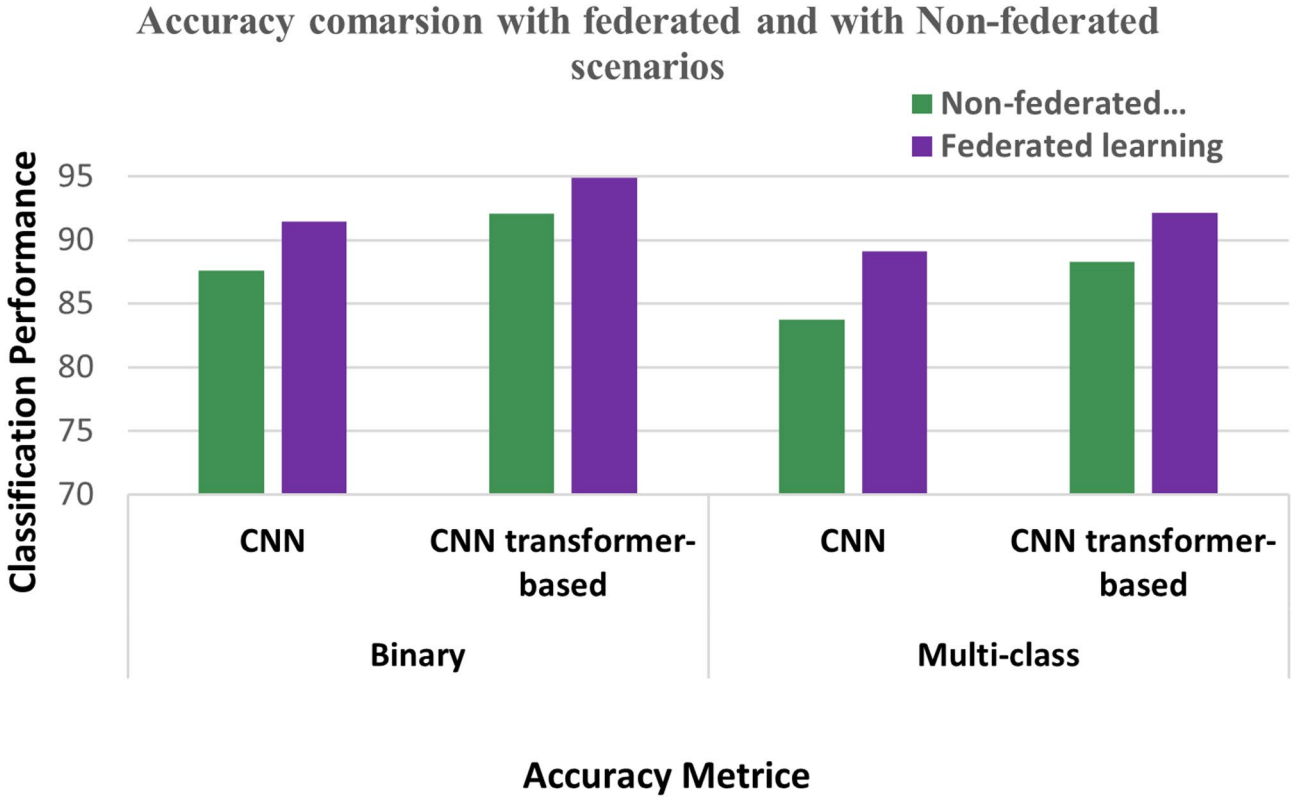


Fig. 7. Comparison the accuracy performance of CNN and the proposed hybrid CNN transformer-based model through both the non-federated scenario and federated learning scenario for binary and multi-class classification approach.

FL vs. centralized

Furthermore, the federated model achieves 94.89% binary classification accuracy, which represents a 2.79% point improvement over the non-federated CNN baseline with 92.10%, and a 0.00% point reduction compared to the non-federated proposed model with 94.89%, demonstrating that the privacy-preserving federated training

introduces no accuracy penalty relative to centralized training while eliminating the need to share raw IoT traffic data across distributed devices.

Moreover, the consistent improvement from the 1st to the 10th round for both models in both classification approaches highlights the effectiveness of federated learning in enabling the models to improve iteratively without requiring centralized data. However, the superior performance of the CNN transformer-based model across all settings suggests that it is better suited for federated learning scenarios, where data may be distributed across multiple clients and communication efficiency is paramount. Figure 7 illustrates the increase in performance achieved by utilizing the transformer with the CNN model when compared to just using CNN for both binary and multi-class classification. Furthermore, it also outlines the methods for implementing the proposed model in a federated learning scenario for improved detection efficiency.

In federated learning, frequent exchange of model parameters between clients and the central server can lead to significant bandwidth consumption, especially in resource-constrained IoT networks. This can be a limiting factor for deploying real-time intrusion detection systems, where timely model updates are crucial. However, to minimize the impact of communication overhead, we employed a limited number of communication rounds (10 rounds), balanced with local training (50 epochs per round) to reduce the frequency of transmissions while maintaining model performance. Additionally, using the FedAvg algorithm helps by transmitting only model weights, rather than raw data, which conserves bandwidth and enhances privacy.

For enhancing security on large-scale systems, implement adaptive client selection, secure aggregation protocols, and differential privacy. In addition, the potential of asynchronous FL frameworks to improve scalability and reduce synchronization delays in large and heterogeneous IoT environments.

Discussion

The study's experimental findings show how well different ML and DL models, including the suggested hybrid CNN-Transformer model, perform in a multi-class classification framework when subjected to various attack types. But there are noticeable differences in performance between classes, which point to areas that need work. In particular, the suggested CNN-Transformer model has a significant F1 score of 87.97% and an overall accuracy of 89.11%. This suggests a robust general ability to categorise network traffic, with the hybrid model showing exceptional performance in detecting DDoS attacks (90.41% F1 score) and PartOfAHorizontalPortScan (99.79% accuracy). These findings imply that the model works well for identifying high-priority threats in Internet of Things systems, which are frequently distinguished by distinct attack patterns. On the other hand, the model performs poorly against C&C and C&C-HeartBeat attacks; its accuracy (14.73%) and F1 score (25.67%) are much lower for C&C. Potential class imbalance may be the cause of this performance discrepancy, as the training data for these classes might not have been accurately representative of actual attack scenarios. Additionally, even though the precision was high (94.63%), the misclassification of benign traffic implies a trade-off between preventing false positives and detecting attacks.

Comparison of time complexity

For the time complexity, Table 13 presents a comparison of the training time complexity (in seconds) for various machine learning and deep learning strategies used in the non-federated learning scenario. The results demonstrate that individual machine learning models, such as XGBoost, RF, and CatBoost, exhibit relatively low training times, with XGBoost being the fastest. The ensemble voting model, which combines predictions from multiple ML models, takes longer due to added complexity. The CNN model incurs a significantly higher time cost due to deep learning computations, while the proposed hybrid CNN-transformer model has the highest training time owing to its integrated architecture of convolutional and self-attention layers. This highlights the trade-off between model complexity and detection accuracy, as the proposed model achieved the highest performance at the cost of greater computational overhead, as discussed in the results section[74].

While centralized IDS may offer advantages in terms of simplicity, unified control, and potentially faster training in controlled environments, it comes with significant limitations in large-scale IoT settings, particularly regarding data privacy, scalability, and single-point failure risks.

Strategy	Models	Time complexity (Seconds)
Individual ML Model	XGBoost	13.40
	Random Forest	71.48
	Catboost	106.45
Ensemble AI	Ensemble voting	212.16
CNN	CNN	747.3
Hybrid AI	Proposed CNN transformer-based	3938.23

Table 13. Comparison of time complexity (in seconds) for different machine learning and deep learning strategies used in the study, including individual models, ensemble methods, and the proposed hybrid CNN-Transformer model.

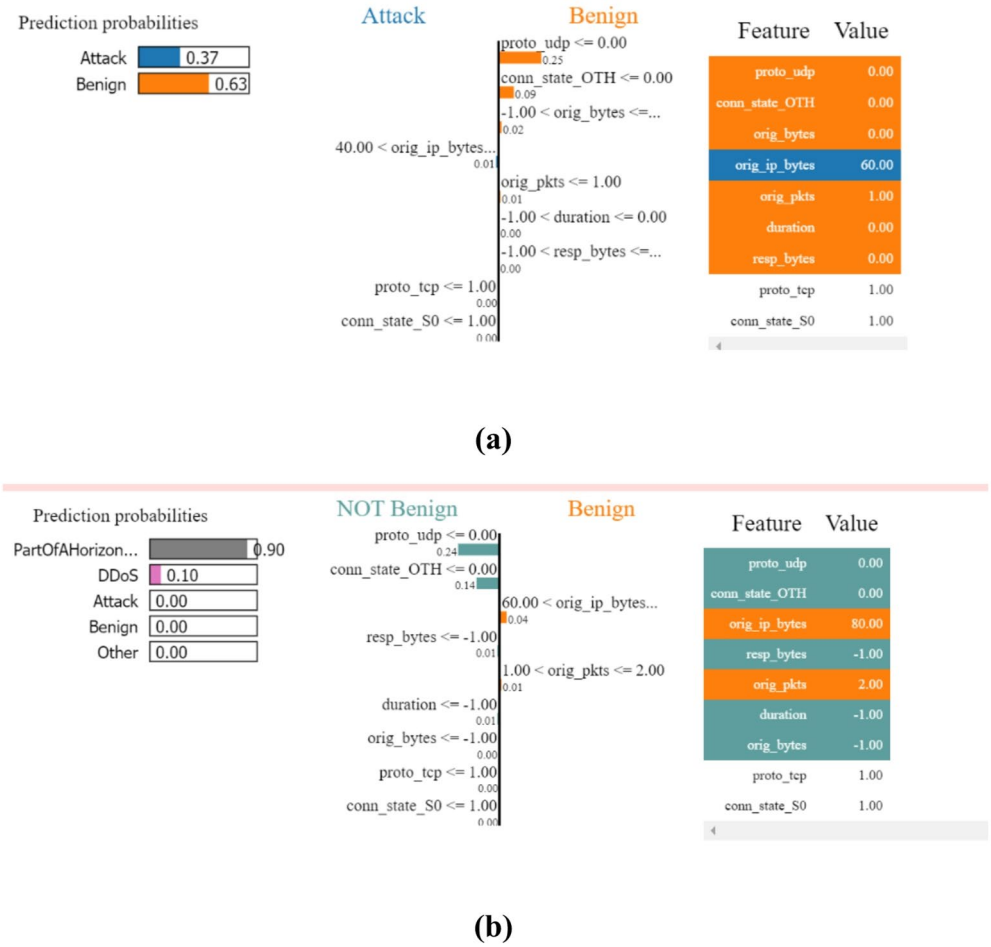


Fig. 8. LIME observations to explain the predictions of the proposed explainable hybrid CNN-transformer model for both approaches (a) the Binary classification, and (b) multi-class classification.

Explainable AI interpretation

In this section, we used the LIME method to explain the predictions of the best classifier models for finding IoT attacks, as shown in Fig. 8 (a, b). This was accomplished by creating a local interpretation of the data using an easy-to-understand model. The model has classified the instance as benign with a probability of 63%, driven by the strong influence of the proto_udp feature and minor contributions from other factors like conn_state_OTH and orig_pkts. The low probability for the attack class (37%) suggests that the features indicative of attacks, such as orig_ip_bytes, contributed only slightly to the attack class prediction. The feature orig_ip_bytes (original IP bytes) with a value greater than 40.00 is contributing slightly to the possibility of an attack, but this contribution is minimal (0.01). The feature proto_udp (UDP protocol) with a value of 0.00 contributes significantly to the model's decision that the instance is benign (0.25). For the multi-class classification, the model classified this instance as PartOfAHorizon with a probability of 0.90, heavily influenced by the absence of UDP traffic (proto_udp = 0) and by certain connection states, such as conn_state_OTH. There is a small probability of 0.10 for DDoS. For the attack, benign, and other classes, the probabilities for these classes are all 0.00, indicating that the model does not predict this instance to belong to these categories. Figure 9 presents the aggregated feature importance across 1000 samples using the LIME technique. This figure visualizes the key features that contribute to the model's decision-making process, providing insights into the most influential factors for predictions across a large dataset. The aggregation of results from 1000 samples offers a more comprehensive understanding of feature importance, helping to identify consistent patterns and trends that can be critical for model interpretation and improvement.

These feature-level attributions provide network security analysts with interpretable, instance-specific evidence to validate or challenge model predictions in real deployments, for example, flagging cases where a high orig_ip_bytes value coincides with a borderline benign classification, warranting manual inspection. This demonstrates that LIME does not merely explain predictions post-hoc, but actively supports human-in-the-loop decision-making in IoT intrusion detection workflows.

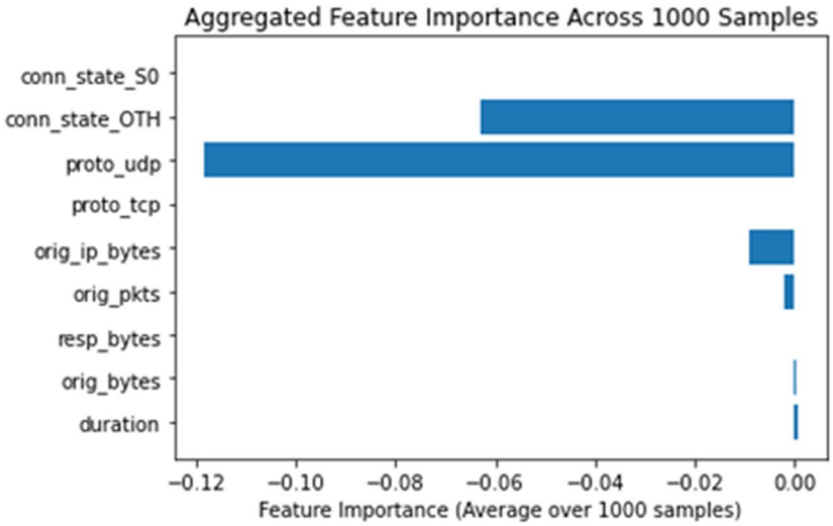


Fig. 9. LIME aggregated feature importance across 1000 samples.

Label with (Encoding number)	Total instances	A training set 70%	A validation set 15%	A testing set 15%
BruteForce (0)	3,619	2542	555	522
DDoS (1)	9,977	6957	1500	1520
web-based attacks DNS (2)	9,987	7027	1454	1506
DoS (3)	7,356	5186	1089	1081
Mirai (4)	5,172	3635	773	764
SqlInjection (5)	6602	4565	1042	995
Spoofing for upload attacks (6)	1348	930	197	221
Total	44,061	30,842	6,610	6,609

Table 14. Data partitioning for training, validation, and testing for multi-class classification approaches on the CIC IoT-DIAD 2024 dataset.

Ablation study

The ablation study employed the CIC IoT-DIAD 2024 dataset to provide the proposed model with generalisation training, a flow-based IoT intrusion detection dataset intended for device identification and anomaly detection[75]. The dataset comprises network traffic data from 105 IoT devices subjected to both benign and malicious conditions, classified into seven principal attack types, including partial attacks such as DDoS, DoS, brute force, spoofing for upload attacks, web-based attacks in DNS, SQL injection, and Mirai botnet attacks. Table 14 provides an overview of the total number of rows for each attack used in the ablation study, including the distribution of data across training, validation, and testing sets.

Figure 10 illustrates the results of feature selection using the RF classifier on the dataset defined above. The classifier was applied to all dataset features, and the feature_importances_ attribute was used to determine each feature's importance score. The diagram shows that the Dst IP and Src IP features have the highest relevance scores, 0.095 and 0.089, respectively. This indicates that they have a significant impact on the model's prediction results. The model also benefits from additional important features, such as the number of RST flags (0.044), the destination port (0.042), and the source port (0.039), as the top 30 features were selected based on their importance scores.

Table 15 presents the performance results of the proposed model for each attack in the multi-class classification job, utilising the CIC IoT-DIAD 2024 dataset. DDoS exhibits the highest accuracy of 98.22% and sensitivity of 98.22%, indicating the model's proficiency in detecting this sort of attack. DoS performs exceptionally, achieving an accuracy of 99.63% and a high F1 score of 99.77%. The SQL Injection attack is 93.77% accurate, 93.77% sensitive, and 95.79% specific. Conversely, the Spoofing for Upload Attacks exhibits a diminished accuracy of 85.52%. The model achieves a commendable F1 score of 88.73% for class 6. The confusion matrix depicted in the illustration further corroborates these findings in Fig. 11. The diagonal elements indicate the number of cases accurately identified for each category. Across all attack types, the model exhibits a significant number of true positives, particularly for DDoS and DoS, which are categorized with the highest accuracy. The negligible off-diagonal elements further substantiate the model's ability to differentiate among various types of attacks.

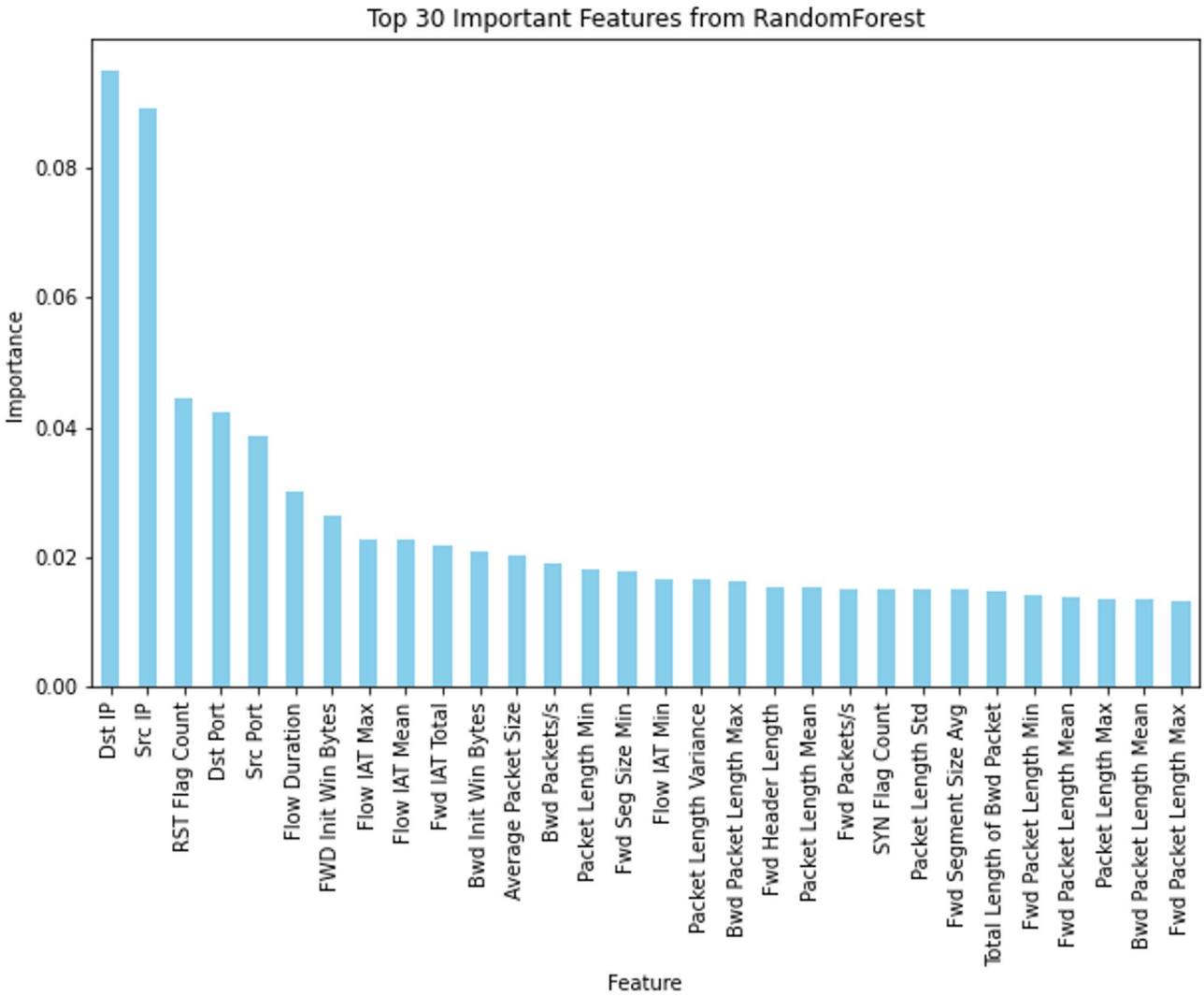


Fig. 10. The random forest method yielded the highest feature significance values.

Class	ACC	REC	PRE	F1
BruteForce (0)	91.95	91.95	89.22	97.05
DDoS (1)	98.22	98.22	99.87	99.04
web-based attacks DNS (2)	99.40	99.40	97.65	98.52
DoS (3)	99.63	99.63	99.91	99.77
Mirai (4)	97.51	97.51	94.78	96.13
SqlInjection (5)	93.77	93.77	95.79	94.77
Spoofing for upload attacks (6)	85.52	85.52	92.20	88.73
Total	97.05	97.05	97.07	97.05

Table 15. The per-class performance result of the proposed model for each of the attack types in the multi-class classification task on the CIC IoT-DIAD 2024 dataset.

Limitations and future works

This study has several limitations that should be acknowledged. First, regarding class imbalance: the IoT-23 dataset exhibits severe class imbalance, with the majority class (PartOfAHorizontalPortScan) containing 753,565 instances compared to only 30 instances in the smallest class (C&C-Torii). Although SMOTE was applied, it produced no measurable improvement due to this extreme disparity. Future work should explore more effective strategies such as: (1) class-weighted loss functions (e.g., focal loss), which penalise misclassification of minority

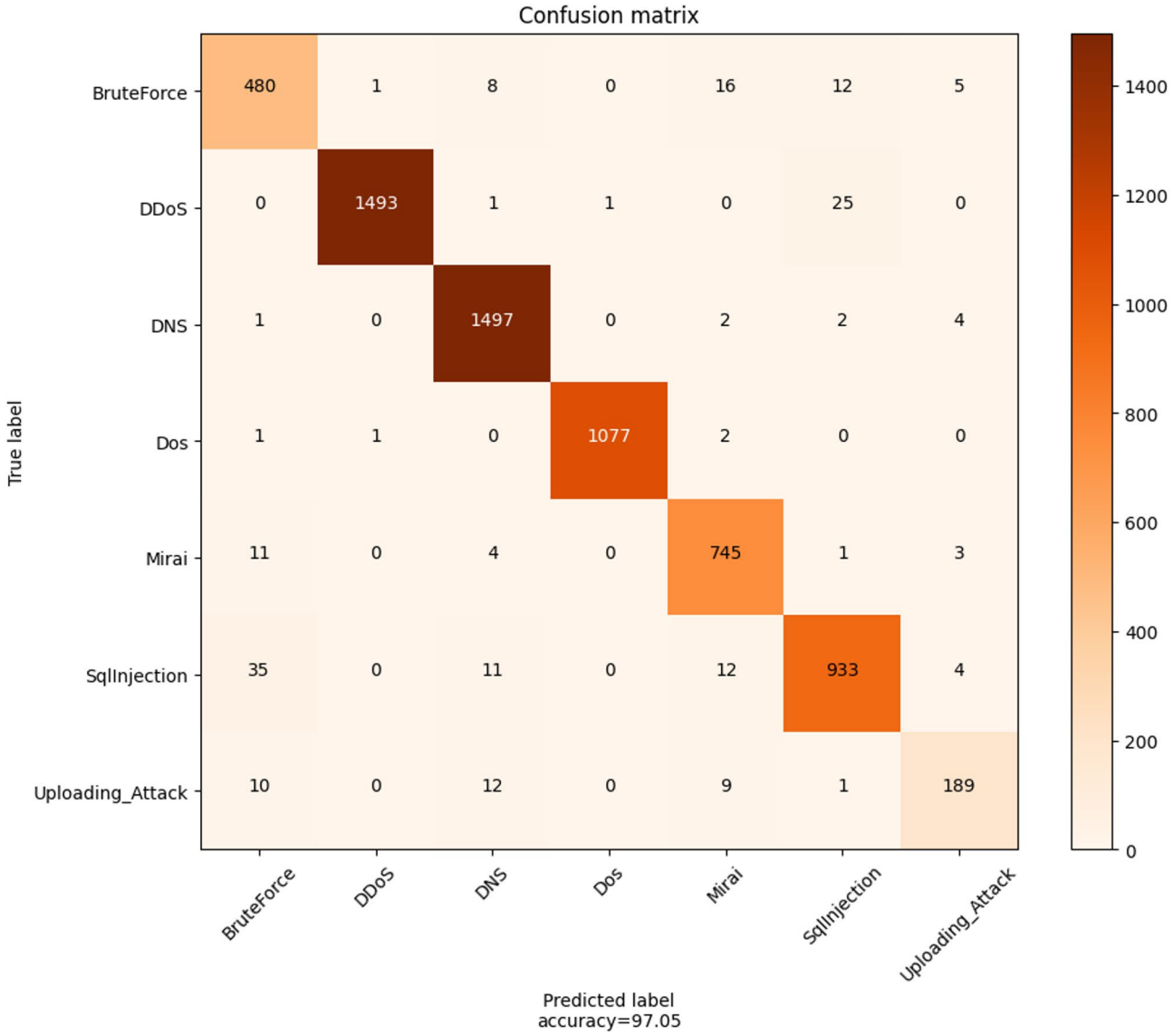


Fig. 11. The confusion matrix of the proposed model on the CIC IoT-DIAD 2024 dataset.

classes more heavily during training; (2) threshold-moving techniques that adjust decision boundaries per class; and (3) hybrid resampling combining SMOTE with Tomek Links to remove majority-class borderline samples. Second, the FL implementation assumes IID data distribution across clients, which may overestimate performance in heterogeneous real-world IoT environments, as discussed in Sect. 3.6. Third, the FL framework faces inherent security risks including model poisoning, where malicious clients deliberately submit manipulated model updates to degrade global performance or introduce backdoors, and inference attacks that may expose private training data. To address these threats, future work will explore Secure Multi-Party Computation (SMPC) within blockchain consensus mechanisms. Fourth, communication overhead between clients and the central server remains a practical constraint in bandwidth-limited IoT networks; future work will investigate adaptive client selection and gradient compression to reduce transmission costs. Future work is to include more diverse datasets from various domains and multimodal forms, such as MBB-IoT[76] with transfer ensemble models[77,78] and transfer ViT models[79,80], and SMPC and work of the stage in consensus mechanism blockchain environments[81].

Communication Cost.

Additionally, communication overhead represents a practical constraint in bandwidth-limited IoT deployments. The estimated communication overhead per federated round is approximately 12 MB, calculated based on the model's approximately 3 million trainable parameters at 4 bytes (float32) per parameter, multiplied across 4 clients. Over 10 communication rounds, the cumulative total communication cost is approximately 480 MB. While this overhead is feasible for IoT gateway devices with standard network connectivity, it represents a practical constraint for severely bandwidth-limited edge nodes. Future work will investigate gradient compression and adaptive client selection strategies to reduce per-round communication cost by an estimated

Reference	Dataset	Labels	Methodology	Accuracy (%)
B. Roy et al. al., 2018[23]	IoT-23	9 labels	LSTM and BRNN	72
N. Liang, et al., 2021[24],	IoT-23	9 labels	CNN, Decision Trees, Naive Bayes, and SVM	74
R. N. Alhamad, et al., 2022[25],	IoT-23	8 labels	RF, CNN	89.22
S. R. Garg, et al., 2022[26]	BoT–IoT, IoT-23	4 labels	XGB and LGB	94.79
A. Nazir et al. 2024[27]	IoT	7 labels	CNN-LSTM	91.99
The proposed model	IoT-23	8 labels	A hybrid CNN-transformer-based mode **in a binary approach**	92.76 in non-federated **94.89 in FL**
			A hybrid CNN-transformer-based mode **in a multi-class approach**	89.11 in non-federated **92.17 in FL**

Table 16. Evaluate the findings concerning the most current research on artificial intelligence for IoT attacks using the IoT-23 dataset.

10–100× without significant accuracy degradation, as demonstrated in the federated learning compression literature.

Comparison evaluation results with the most recent research studies

Using the IoT-23 dataset, this section assesses the suggested model's performance by contrasting it with other recent research on IoT attack classification. Since accuracy is the most popular performance metric for detecting IoT attacks, it serves as the basis for the comparison. The comparison's findings are compiled in Table 16. With an accuracy of 92.76% in a binary classification approach and 89.11% in a multi-class setup, the suggested model performs competitively with state-of-the-art techniques. The accuracy increases to 94.89% and 92.17%, respectively, when FL is used. Chosen studies that used comparable datasets (IoT-23) and concentrated on a range of machine learning approaches, such as LSTM, CNN, SVM, and ensemble methods, for a more thorough comparison.

Conclusion

This study developed a CNN-transformer-based model capable of elucidating its methodology for detecting IoT attacks in both federated and non-federated learning environments. The integration of transformers and CNNs has demonstrated significant potential in identifying intricate IoT attack patterns, achieving higher accuracy while requiring more computational resources related to the base CNN. The federated learning framework enables model training across several clients without compromising privacy, rendering it suitable for extensive IoT networks. Comprehensive testing has demonstrated the model's accuracy in identifying various types of attacks. The model outperforms others, achieving 92.10% accuracy in non-federated binary classification and 94.89% accuracy in federated binary classification. It achieves 89.11% accuracy in non-federated classification and 92.17% accuracy in federated learning for multi-class classification, respectively. The integrated explainability feature fosters trust by allowing users to understand the model's decision-making process. The findings indicate that this strategy significantly enhances the security of IoT ecosystems by providing precise, scalable attack detection tools that safeguard privacy. Furthermore, our use of explainability enables stakeholders to scrutinize the rationale behind identified abnormalities, hence enhancing the transparency and integrity of security solutions for IoT environments.

Data availability

The database is publicly accessible at the following link: [https://www.stratosphereips.org/datasets-iot23] (https:/www.stratosphereips.org/datasets-iot23).

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Competing interests
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Informed consent
Informed consent was not required, as the study used the IoT-23 dataset, which is publicly accessible data with no identifiable information about individuals.

Additional information
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